



CLIK-Diffusion: Clinical Knowledge-informed Diffusion Model for Tooth Alignment

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ABSTRACT

Traditional semi-automatic methods for tooth alignment involve laborious manual procedures and heavily depend on the expertise of dentists, which often leads to inefficient and prolonged treatment durations. Although many automatic methods have been proposed to assist especially the less experienced dentists, they often lack incorporating clinical insight and oversimplify the problem by estimating rigid transformation matrix for each tooth directly from dental point clouds. This over-simplification fails to capture nuanced requirements of orthodontic treatment, i.e., specific clinical rules for effective alignment of misaligned teeth. To address this, we propose CLIK-Diffusion, a Clinical Knowledge-informed Diffusion model for automatic tooth alignment. CLIK-Diffusion formulates the complex problem of tooth alignment as a more manageable landmark transformation problem, which is further refined into a landmark coordinate generation task. Specifically, we first detect landmarks for each tooth by category, and then build our CLIK-Diffusion to learn distribution of normal occlusion. To further encourage the integration of essential clinical knowledge, we design hierarchical constraints from three perspectives: (1) dental-arch level: to constrain arrangement of teeth from a global level; (2) inter-tooth level: to ensure tight contact and avoid unnecessary collision between neighboring teeth; and (3) individual-tooth level: to guarantee correct orientation of each tooth. In this way, our designed CLIK-Diffusion is able to predict the post-orthodontic landmarks that align with clinical knowledge, and then estimate rigid transformation for each tooth based on coordinates of its pre- and post-orthodontic landmarks. We have evaluated our CLIK-Diffusion on various malocclusion cases collected in real-world clinics, and demonstrate its exceptional performance and strong applicability in orthodontic treatment, compared with other state-of-the-art methods. Our dataset and code is available at <https://github.com/ShanghaiTech-IMPACT/CLIK-Diffusion>.

1. Introduction

Orthodontic treatment, aimed at rectifying misaligned teeth (i.e., malocclusion), offers an effective solution to malocclusion problems. It is estimated that over 56% of people suffer from malocclusion problems with various degrees (Stomatologic, 2020), and most people can benefit from orthodontic intervention. This treatment not only enhances patients' confidence and psychological well-being, but also safeguards patients' oral health and prevents oral diseases (Grewal et al., 2019; Geiger, 2001). Tooth alignment, an essential stage in orthodontic treatment, aims to anticipate the ideal dentition from a set

of misaligned teeth. However, the traditional workflow involves tedious and laborious manual procedures that heavily depend on expertise of dentists, leading to inefficient and prolonged treatment durations.

With rapid development of digital technologies, several computer-aided methods have been proposed for automatic orthodontic treatment, devoted to addressing the aforementioned challenges. Lingchen et al. (2020) proposed a lightweight model called TALigNet; however, it simply uses a MLP-based network to regress each tooth's pose with trivial results. Wei et al. (2020) proposed a tooth arrangement network based on PointNet (Qi et al., 2017) and graph neural network,

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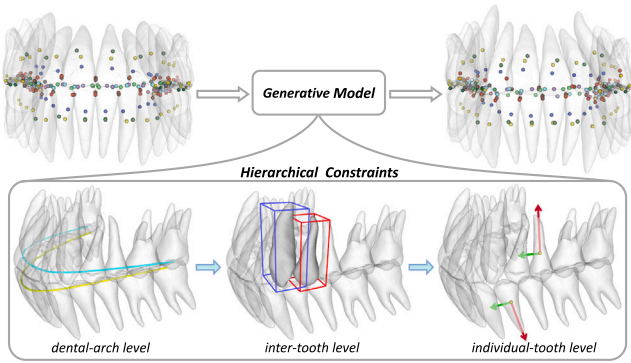


Fig. 1. Overview of our formulation. We formulate the complex tooth alignment task as a more manageable landmark generation task. This is facilitated by a generative model which is guided by hierarchical clinically-inspired constraints.

called TANet, aiming to directly regress the transformation matrices of teeth. Lei et al. (2024) introduced a U-ViT (i.e., a vision transformer-based diffusion model Bao et al., 2023), called TADPM, to predict 6-DOF rotation and 3D translation for each individual tooth, leveraging global features from full-dentition point clouds and local features from tooth meshes by MeshMAE (Liang et al., 2022). Deng et al. (2024) proposed a framework called TAPoseNet that employs a DGCNN (Wang et al., 2019) to estimate the initial poses of teeth, and ultimately uses an MLP to regress the target pose for each individual tooth. However, none of these methods incorporates clinical prior knowledge, but simply learns the mapping relationships between two states, which leads to limited clinical applicability of results. In terms of integrating clinical knowledge into tooth alignment, the closest work to ours is TANet+ (Wang et al., 2022), which is an extended work based on their previous TANet. Specifically, TANet+ maintains a similar network structure but introduces tooth landmarks as fine-grained constraints to make teeth normally inclined. Nonetheless, current methods still suffer from two major limitations: (1) point cloud-based methods (e.g., PointNet or DGCNN) or mesh-based methods (e.g., MeshMAE) naturally struggle to capture fine-grained local structures and spatial relationships of teeth, and (2) alignment results do not adequately align with clinical knowledge well enough.

To address the mentioned limitations, in this paper, we propose CLIK-Diffusion, a *CL*inical *K*nowledge-informed *D*iffusion model for automatic tooth alignment. First of all, we introduce a new formulation for this tooth alignment task. Instead of using point cloud representation with PointNet for a transformation matrix regression task, we introduce tooth landmark representation with a generative model for a landmark generation task as shown in Fig. 1. Tooth landmarks, predetermined by expert dentists, can significantly represent the position, orientation, and shape of teeth throughout the orthodontic process. Since our generative model is founded on tooth landmark representation, it creates a natural pathway for integrating essential clinical knowledge on the level of landmarks. Inspired by the empirical clinical rules of tooth alignments, namely Andrews' Six Keys & Six Elements (Andrews, 1972; LF, 2000), which are crucial criterion for achieving proper occlusion and ensuring successful orthodontic treatment (see Section 2.3), we design hierarchical constraints from three perspectives: (1) dental-arch level: to constrain arrangement of teeth from a global level; (2) inter-tooth level: to ensure tight contact and avoid unnecessary collision between neighboring teeth; and (3) individual-tooth level: to guarantee correct orientation of each tooth. In this way, our CLIK-Diffusion is able to predict the post-orthodontic landmarks that align with clinical knowledge. Once the post-orthodontic tooth landmarks are predicted, rigid transformations for each tooth can be determined using the Kabsch algorithm (Kabsch, 1976), combined with its pre-orthodontic tooth landmarks. These optimal transformations can then be applied to each tooth to generate the

post-orthodontic dentition. Our proposed method has been validated on various malocclusion cases collected in real-world clinics and the experimental results demonstrate our method achieves superior performance compared to the state-of-the-art methods. The contributions of this paper are summarized as follows:

1. We propose CLIK-Diffusion, the first generative model for tooth alignment, which introduces an innovative formulation, namely landmark representation, enabling spatial- and shape-aware tooth arrangement.
2. Our CLIK-Diffusion incorporates hierarchical constraints from three perspectives: dental-arch level, inter-tooth level, and individual-tooth level, which are inspired by empirical rules in clinical practice.
3. Extensive evaluations, including comparative experiments, external validations, ablation studies, and user studies, demonstrate that our method can predict well-aligned results, delivering superior qualitative and quantitative performance compared to existing methods. This highlights the effectiveness and clinical capability of our proposed CLIK-Diffusion.

2. Related work

2.1. Digital dentistry

Digital Dentistry, which refers to the use of digital or computer-aided technologies to carry out dental procedures, was pioneered by Francois Duret, who created the first CAD/CAM system for dentistry in the 1970s (Tout-C ramique, 1971; Duret et al., 1988). Since then, numerous advanced technologies have been incorporated into the field of dentistry. For instance, early works focused on utilizing neural networks to conduct tooth segmentation on CBCT images or intraoral scanings (Cui et al., 2019, 2022; Zhao et al., 2021). Moreover, recent advancements have also been made in detecting dental caries on images (Lee et al., 2018; Imak et al., 2022). Additionally, deep learning techniques are applied to the diagnosis of periodontal diseases, improving the accuracy of identifying periodontitis (Chang et al., 2020, 2022). In the field of dental implant, many AI-driven approaches have been developed to assist in planning the placement of dental implants, ensuring better outcomes and reducing surgical risks (Kurt Bayrakdar et al., 2021; Park et al., 2022). These innovations collectively contribute to the evolution of digital dentistry, enhancing both diagnostic capabilities and treatment efficacy.

2.2. Tooth alignment

Tooth alignment is an essential stage in orthodontic treatment for rectifying misaligned teeth. Given an ill-positioned dentition of a patient, tooth alignment aims to predict an ideal tooth arrangement, guiding the design and fabrication of customized orthodontic appliances (Tarraf and Ali, 2018). Traditional tooth alignment methods rely on manual, mechanically repetitive operations, where professional dentists typically design treatments using 3D models scanned from plaster casts of patients' teeth (Fleming et al., 2011). While specialized orthodontic systems, such as OrthoAnalyzer[ ] (Thomas and Sivakumar, 2023) and OrthoCAD[ ] (Joffe, 2004), offer some assistance to dentists to some extent (Westerlund et al., 2015), the process remains tedious and time-intensive.

To address the aforementioned challenges, several learning-based methods have recently been proposed to automatically generate an ideal dentition, building on the rapid progress in digital technologies. Wei et al. (2020) introduced TANet, the first learning-based approach for predicting an optimal dentition, given the pre-orthodontic misaligned dentition. TANet employs PointNet to extract both jaw-level and tooth-level features from tooth point clouds and utilizes a graph neural network to implement a feature propagation module for

capturing topological relations. TAlignNet, presented by Lingchen et al. (2020), utilizes a PointNet encoder to extract a unique ‘geometry code’ for each tooth. This code, combined with the tooth’s initial pose, is fed into an MLP-based decoder to predict the target pose. Wang et al. (2022) proposed TANet+, which is an extended work based on their previous work TANet. While maintaining a similar network structure, TANet+ introduces landmarks to constrain the orientation of each individual tooth. TADPM, proposed by Lei et al. (2024), employs a vision transformer-based diffusion model to predict transformations for teeth, combined with features captured from tooth point clouds and tooth meshes. Deng et al. (2024) introduced TAPoseNet, with the network architecture paralleling earlier methods such as TANet and TANet+, which predicts target tooth poses directly rather than estimating transformations from initial states. However, existing methods either fail to fully incorporate clinical knowledge or do not incorporate it at all.

Besides, Fan et al. (2024) proposed a collaborative tooth motion diffusion model, aimed to generate an optimal sequence of tooth movements from the initial dentition to the given target dentition. While pioneering the application of diffusion models in orthodontics, their task fundamentally differs from ours, which is to predict the target dentition directly from the initial state. Our task serves as a prerequisite for the resolution of their task.

2.3. Clinical knowledge in tooth alignment

In clinical practice, the procedure for tooth alignment strictly adheres to certain empirical rules, particularly the **Andrews’ Six Keys & Six Elements**. The Andrews’ Six Elements (LF, 2000), which are derived on the basis of Andrews’ Six Keys (Andrews, 1972), both serve as the most mainstream principles for tooth alignment task in the field of orthodontics. Such empirical clinical rules are the cornerstone of achieving orofacial harmony for optimal health and aesthetics, which are vital for ensuring effective orthodontic treatments. These essential clinical knowledge has been summarized as hierarchical constraints at three levels: the global dental-arch level, the inter-tooth level, and the local individual-tooth level. (1) Specifically, at the most global level (i.e., dental-arch level), Andrews’ Six Elements focus on determining the ideal position of teeth and morphological characteristics of ideal dental arch, known as the WALA ridge (Triviño et al., 2010). By leveraging the uniqueness of dental arch to guide tooth alignment, each patient can receive a personalized orthodontic treatment, gaining a well-aligned dentition. (2) Drilling down to a more localized perspective (i.e., inter-tooth level), Andrews’ Six Keys emphasize the critical role of contact relationships between neighboring teeth. An ideal dentition is supposed to ensure tight contact between neighboring teeth while mitigating unnecessary collision, thereby promoting smoother and more efficient tooth movement. (3) Finally, at the most fine-grained level (i.e., individual-tooth level), Andrews’ Six Keys concentrate on optimizing orientation of each tooth, ensuring that every tooth achieves an ideal angulation and inclination within the dental arch.

3. Method

3.1. Problem formulation

Fig. 2 illustrates our algorithm pipeline. The set of initial tooth models is symbolized as $X_{initial} = \{X_1, X_2, \dots, X_N\}$, where $N = 28$ represents the total number of teeth for a single patient. With these 28 pre-orthodontic tooth models within a single dentition, we aim to predict the corresponding 28 post-orthodontic tooth models $\hat{X}_{target} = \{\hat{X}_1, \hat{X}_2, \dots, \hat{X}_N\}$, which would emulate the well-aligned teeth designed by experienced dentists.

As mentioned in Section 2.2, existing methods adhere to traditional formulation that directly regresses transformation matrices from

extracted features. This approach necessitates the use of point cloud-based encoders for feature extraction, which often leads to loss of spatial information related to the teeth. Furthermore, integrating clinical knowledge into the transformation regression process continues to pose a significant challenge.

Hence, in this paper, we propose a new formulation that converts the tooth alignment task into a more manageable task of landmark coordinate generation by the powerful diffusion model. Tooth landmarks, being more concise, are position- and orientation-aware representations, thereby improving efficiency and better fitting our task. To this end, we first detect a few depictive landmarks for each tooth using a lightweight neural network (see Section 3.2.1 and Fig. 2 left). It takes $X_{initial}$ as input and predicts $L_{initial} = \{L_1, L_2, \dots, L_N\}$, where $L_i \in L_{initial}$ is the total landmarks of one tooth. Subsequently, we sample a fixed number of points surrounding landmarks for each individual tooth mesh to form the dentition descriptor, aimed to capture more shape-aware features (Section 3.2.2 and Fig. 2 left). These landmarks and the dentition descriptor together serve as condition of our diffusion model, for which, we have designed hierarchical constraints inspired by clinical knowledge to supervise the diffusion training (Section 3.3 and Fig. 2 middle). Our diffusion model outputs the desired target landmarks $\hat{L}_{target} = \{\hat{L}_1, \hat{L}_2, \dots, \hat{L}_N\}$. Finally, with $L_{initial}$ and $\hat{L}_{target}$, we vote for a single transformation for each tooth with the help of least square optimization and transform $X_{initial}$ to produce $\hat{X}_{target}$ (Section 3.4 and Fig. 2 top right).

3.2. Preliminary

3.2.1. Landmark detection

As defined by expert dentists, different types of teeth possess distinct numbers and types of landmarks. As shown in Fig. 3, incisors and canines have the same number of landmarks (i.e., 8) while premolars have 9, and molars have 11. These landmarks include centroids, cusps, facial axis points, as well as contact points, and are distributed differently. For a dentition with $N = 28$ teeth, we obtain a total of 256 landmarks.

To faithfully detect all landmarks from the pre-orthodontic dentition $X_{initial}$, we develop a landmark detection network based on PointMLP (Ma et al., 2022; Wei et al., 2022). Specifically, it utilizes residual MLP blocks to progressively extract features from the input tooth point clouds (each tooth with 2048 points sampled from tooth mesh), thereby gaining a deeper understanding of the structural and geometric information of each tooth. To stabilize the prediction and reduce ambiguity in landmark labels from manual annotation, our network outputs per-landmark heatmaps, instead of directly regressing landmark coordinates. These generated heatmaps are compared against ground truth heatmaps, converted from ground truth landmarks in our dataset, using a simple L_2 loss following most previous works (Zhang et al., 2020; Wu et al., 2024).

After training, we simply apply the Argmax operator to predict landmark coordinates $L_{initial} \in \mathbb{R}^{256 \times 3}$ from the initial dentition. For better guidance, we concatenate these pre-orthodontic landmark coordinates $L_{initial}$ with their respective landmark IDs ($\mathbb{R}^{256 \times 1}$) as well as tooth IDs ($\mathbb{R}^{256 \times 1}$) to form a strong condition $L_c \in \mathbb{R}^{256 \times 5}$, where each landmark knows its position, as well as the tooth it belongs to. Note that, we use zero padding to handle cases with missing teeth. Additionally, we can readily obtain landmark coordinates of the target dentition (i.e., post-orthodontic landmarks $L_{target} \in \mathbb{R}^{256 \times 3}$), given that we possess the ground truth transformation matrix for each individual tooth. L_{target} will serve as the ground truth to supervise our model during training process.

3.2.2. Dentition descriptor

Having established a comprehensive set of 256 tooth landmarks following Section 3.2.1, which can represent the position and orientation of various teeth, we can delve further into describing the shape features of each tooth within the whole dentition. For each landmark

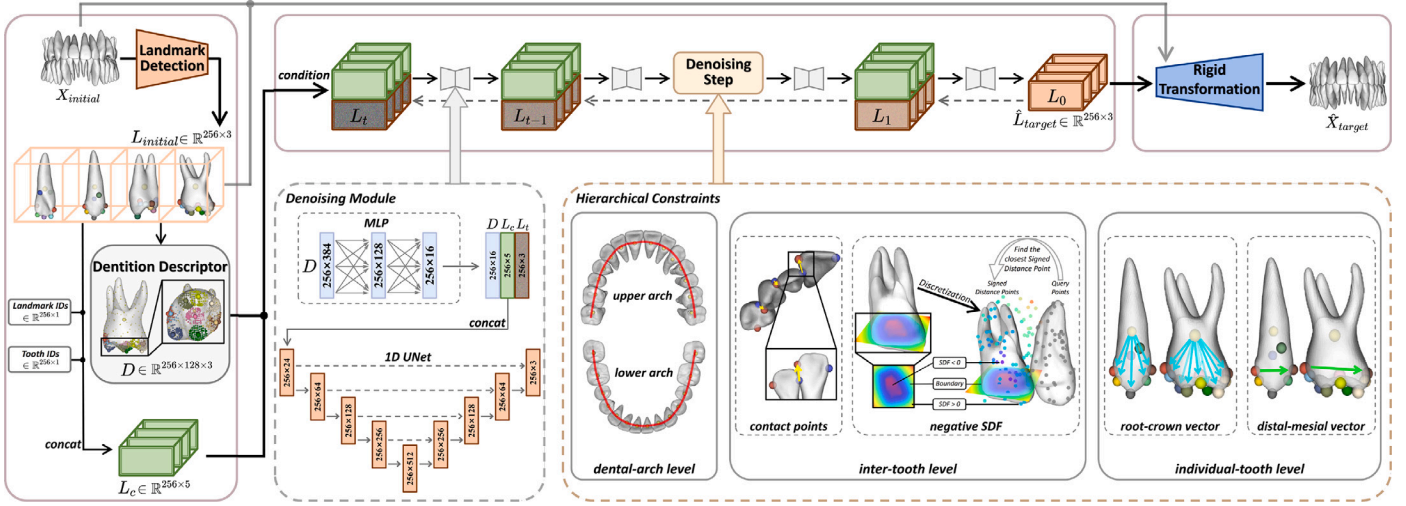


Fig. 2. Overview of our proposed CLIK-Diffusion. Given the pre-orthodontic dentition $X_{initial}$, we first detect all tooth landmarks and then sample points surrounding landmarks (Section 3.2), which then serve as condition for our diffusion model to predict the desired target landmarks (Section 3.3). Finally, we derive rigid transformations (Section 3.4) for each tooth from the initial and predicted landmarks and apply them to $X_{initial}$ to obtain the post-orthodontic dentition $\hat{X}_{target}$. Note that the left dashed box depicts the detailed structure of our denoising module within the diffusion model, while the right dashed box illustrates the calculation process of hierarchical constraints.

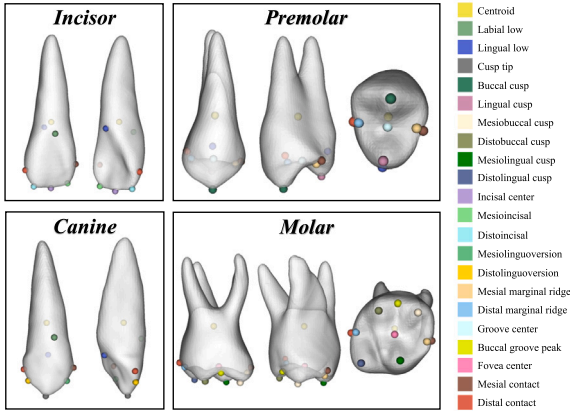


Fig. 3. Illustration of tooth landmarks for different types of teeth as defined by Andrews' Six Keys and dental experts. The landmarks are color-coded and labeled with their medical terms.

of a tooth excluding the centroid, we sample 128 vertices surrounding it by defining a sphere with a radius of 2 millimeters centered at the landmark (see Fig. 2 left). However, when it comes to the centroid of each tooth, we adopt a different technique that utilizes a Farthest Point Sampling (FPS) method to sample 128 vertices from across the entire tooth mesh, providing a global portrayal of the tooth's shape and geometry. Consequently, following the aforementioned description, after performing vertex sampling for all 256 landmarks, we obtain the dentition descriptor, denoted as $D \in \mathbb{R}^{256 \times 128 \times 3}$, where 128 represents the number of sampled vertices and 3 represents the 3D coordinates of each sampled vertex. This dentition descriptor is then flattened to $\mathbb{R}^{256 \times 384}$. This approach of dentition descriptor allows us to gather detailed information about each tooth at specific anatomical locations, enhancing our understanding of its shape features.

3.3. Clinical knowledge-informed constraints

Given the landmark condition L_c and the dentition descriptor D as guidance, we introduce a diffusion model with clinical knowledge-informed constraints to generate the coordinates of our desired landmarks $\hat{L}_{target} \in \mathbb{R}^{256 \times 3}$ of the target dentition, with the supervision of the ground truth coordinates of target landmarks L_{target} .

We formulate this step as a diffusion process: a forward noise-adding process that gradually adds noise to the landmarks, guiding them towards a Gaussian distribution ($\mathbb{R}^{256 \times 3}$), followed by a reverse denoising process that progressively removes noise from these noisy landmarks. The forward noise-adding process is modeled as a Markov process using:

$$q(L_t | L_{t-1}) = \mathcal{N}(L_t; \sqrt{\alpha_t} L_{t-1}, (1 - \alpha_t) \mathbf{I}), \quad (1)$$

where t is the noising time step, the constant $\alpha_t \in (0, 1)$ is a hyper-parameter for noising, and L_t represents the noise landmarks at time step t , same for L_{t-1} . Note that there is no training in the forward noise-adding process. During the reverse denoising process at time step t , we have:

$$p_\theta(L_{t-1} | L_t) = \mathcal{N}(L_{t-1}; \hat{\mu}_\theta(L_t, L_c, D, t), \hat{\beta}_t \mathbf{I}), \quad (2)$$

where p_θ denotes the denoising module of our diffusion model, $\hat{\mu}_\theta(\cdot)$ is the parameter to be learned during diffusion process and $\hat{\beta}_t$ is a hyper-parameter correlated to α_t .

As illustrated in Fig. 2 bottom, our denoising module comprises two main components. (1) Initially, a three-layer MLP extracts features from the dentition descriptor D , transforming it into a $\mathbb{R}^{256 \times 16}$ dimensional representation. (2) Subsequently, the 1D UNet component processes a concatenated input of $\mathbb{R}^{256 \times 24}$, consisting of $L_t \in \mathbb{R}^{256 \times 3}$, $L_c \in \mathbb{R}^{256 \times 5}$, and the processed $D \in \mathbb{R}^{256 \times 16}$. Inspired by Tevet et al. (2023), at any time step t , we directly predict the clean target landmarks $\hat{L}_{target}(t)$ instead of predicting the noise that should be removed. We then add noise on $\hat{L}_{target}(t)$ to derive L_{t-1} at the previous time step. This approach enables us to apply several clinical knowledge-informed constraints defined on landmarks. After all time steps are completed, our denoising module can generate the landmarks at time step $t = 0$ (i.e., $L_{t=0}$), which corresponds to the output of our network, namely our final predicted target landmark $\hat{L}_{target}$.

Regarding the training objective, we primarily employ the inherent loss function of diffusion model, denoted as $\mathcal{L}_{Base}$. $\mathcal{L}_{Base}$ is calculated using the L_2 distance between the denoised landmark coordinates and the ground truth coordinates, aimed for constraining our model to learn the distribution of dentitions with normal occlusion. The mathematical formula is defined as:

$$\mathcal{L}_{Base} := \mathbb{E}_{t \sim [1, T]} \left[\left\| L_{target} - \hat{L}_{target}(t) \right\|_2^2 \right], \quad (3)$$

where L_{target} is the ground truth landmarks, $\hat{L}_{target}(t)$ is the target landmarks predicted at any time step t , and T is the total denoising time steps.

In addition to the inherent loss described above, for fully leveraging clinical knowledge in orthodontic treatment, we implement hierarchical constraints (from dental-arch level, inter-tooth level to individual-tooth level), as described below.

3.3.1. Dental-arch level constraint

To constrain the arrangement of teeth from the global dental-arch level, we aim for the target teeth to closely match the ground truth teeth in terms of the characteristics of dental arch. For both the upper and lower teeth, we utilize a fourth-order polynomial to characterize the morphological features of dental arch. The fourth-order polynomial, determined by five coefficients, is a widely-used curve in clinical orthodontics, providing a more flexible and smooth representation of the dental arch (AlHarbi et al., 2008; Adaškevičius and Vasiliauskas, 2009). As shown in Fig. 2, taking the upper teeth as an example (the same for the lower teeth), we first project the landmarks onto the horizontal plane (i.e., ignoring the z-axis). Then we employ a differentiable approach by fitting a fourth-order polynomial to the average coordinates of all landmarks for each tooth, which represent its morphological center. This allows us to obtain five coefficients that form a five-dimensional vector. Subsequently, we utilize the L_2 loss to align the coefficients of our predicted dental arch with those of the ground truth, thereby controlling both the size and shape of dental arch. Then, the dental-arch level loss is defined as:

$$\mathcal{L}_{Arch}^{u/l} := \|F(L_{target}^{u/l}) - F(\hat{L}_{target}^{u/l})\|_2^2, \quad (4)$$

$$\mathcal{L}_{Arch} := \mathcal{L}_{Arch}^u + \mathcal{L}_{Arch}^l, \quad (5)$$

where the superscript u/l denotes that we fit curves for the upper and lower teeth, respectively, $F(\cdot)$ represents the process of determining the five coefficients of the fourth-order polynomial that fits the dental arch.

3.3.2. Inter-tooth level constraint

The contact relationships between neighboring teeth are crucial considerations in clinical orthodontic treatment. In an ideal dental arch, the distance between neighboring teeth should be minimized to achieve close contact while avoiding unnecessary collision. As depicted in Fig. 3, each tooth has landmarks defined by dentists which include two contact points, representing the regions where it touches neighboring teeth on both sides. Thus, we can make neighboring teeth closer by minimizing the L_2 distance between the contact points of each pair of neighboring teeth. We introduce the contact loss to enforce this constraint:

$$\mathcal{L}_{Contact}^{u/l} := \frac{1}{13} \sum_{i=1}^{13} \|\hat{r}_{contact,i}^{right} - \hat{r}_{contact,i+1}^{left}\|_2^2, \quad (6)$$

$$\mathcal{L}_{Contact} := \mathcal{L}_{Contact}^u + \mathcal{L}_{Contact}^l, \quad (7)$$

where $\hat{r}_{contact,i}^{right}$ is the predicted right contact point of the i -th tooth and $\hat{r}_{contact,i+1}^{left}$ is the predicted left contact point of the $(i+1)$ -th tooth, assuming the tooth ID ascends from left to right for both the upper and lower teeth. Similarly, the superscript u/l means the upper and lower teeth. Note that the number 13 indicates the total pairs of contact points for either the upper or lower teeth, as the 14th tooth lacks a right neighboring tooth and thus is not paired.

Meanwhile, to mitigate unnecessary collision between neighboring teeth, we introduce an additional constraint based on the Signed Distance Field (i.e., SDF) (Park et al., 2019). SDF is commonly used to describe the spatial relationship between points in space and a given object: a negative SDF value indicates that the point lies inside the object, while a positive value indicates it is outside. However, directly deforming the field along with the rigid transformations of teeth is challenging. To address this, we discretize the signed distance field into a set of signed distance points. Specifically, for each of the 28 teeth, we randomly sample points in space and compute their corresponding

SDF values. These points, referred as signed distance points, densely cover the interior, surface, and exterior of each tooth. Additionally, a fixed number of points are sampled directly on the tooth surface, which we refer to as query points. During training, given the initial landmarks $L_{initial}$, we predict the target landmarks $\hat{L}_{target}$, thereby deriving transformation matrix for each tooth in a differentiable manner. These transformations are applied to both the signed distance points and the query points, resulting in their updated positions.

For each query point on a given tooth surface, we use a differentiable Soft-Argmin operator (Zhong et al., 2017) to search for the closest signed distance point on both sides of neighboring teeth and assign its SDF value to the query point. If the assigned SDF value is negative, it indicates that the closest point of the query point lies inside a neighboring tooth, signifying a collision between the teeth. To quantify this, we sum the negative SDF values for all query points across each tooth and then average the total across the 28 teeth. This leads to our collision loss function:

$$\mathcal{L}_{Collision} := \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^{M_i} (\min(\text{SDF}(q_{i,j}), 0))^2, \quad (8)$$

where N is the total number of teeth, M_i is the number of query points for the i th tooth, and $\text{SDF}(q_{i,j})$ is the SDF value assigned to the j th query point of the i th tooth. The minimum function ensures that only query points with negative SDF values contribute to the loss.

We formulate the overall loss to constrain the contact relationships at the inter-tooth level:

$$\mathcal{L}_{Inter} := \mathcal{L}_{Contact} + \lambda_4 \mathcal{L}_{Collision}, \quad (9)$$

where λ_4 is used to counterbalance the gradient contributions introduced by the weighted Soft-Argmin operator.

3.3.3. Individual-tooth level constraint

For each individual tooth, landmarks can effectively embody its pose, which can be used to determine whether it is in the correct orientation. The root–crown direction (i.e., the direction from tooth root to tooth crown) and distal–mesial direction (i.e., the tangent direction along the dental arch) are crucial in clinical orthodontics. Applying constraints from these two directions ensures that each tooth is closely aligned with its ground truth orientation. Hence, we introduce the individual-level loss from root–crown and distal–mesial directions as:

$$\mathcal{L}_{Individual} := \frac{1}{N} \sum_{i=1}^N (1 - S_c(\mathcal{V}_1(L_{target,i}), \mathcal{V}_1(\hat{L}_{target,i}))) + \frac{1}{N} \sum_{i=1}^N (1 - S_c(\mathcal{V}_2(L_{target,i}), \mathcal{V}_2(\hat{L}_{target,i}))), \quad (10)$$

where $L_{target,i}$ is the ground truth landmarks of the i th tooth, $\hat{L}_{target,i}$ represents the predicted landmarks of the i th tooth. $\mathcal{V}_1(\cdot)$ calculates the average root–crown vector of one tooth as $\mathcal{V}_1 = \frac{1}{S} \sum_{j=1}^S (l_{crown}^j - l_{centroid}^j)$, where $l_{centroid}$ denotes the centroid of each tooth, l_{crown}^j represents the j th landmark within the tooth crown, and S is the total number of landmarks within the tooth crown. $\mathcal{V}_2(\cdot)$ computes the distal–mesial vector of one tooth as $\mathcal{V}_2 = l_{contact}^{right} - l_{contact}^{left}$, where $l_{contact}^{right}$ and $l_{contact}^{left}$ are the two contact points of one individual tooth. S_c is the cosine similarity.

Finally, we combine the inherent constraint of diffusion model with three hierarchical clinical-inspired constraints to train the diffusion process:

$$\mathcal{L}_{Joint} := \mathcal{L}_{Base} + \lambda_1 \mathcal{L}_{Arch} + \lambda_2 \mathcal{L}_{Inter} + \lambda_3 \mathcal{L}_{Individual}. \quad (11)$$

3.4. Rigid transformation

Given the pre-orthodontic tooth landmarks $L_{initial}$, once we predict the post-orthodontic landmarks $\hat{L}_{target}$, we utilize Kabsch algorithm (Kabsch, 1976) to obtain the per-tooth rigid transformation

between pre- and post-orthodontic landmarks. These transformations can then be applied to each 3D tooth model within the initial dentition $X_{initial}$, to produce the post-orthodontic dentition $\hat{X}_{target}$, which can be regarded as the outcome of the orthodontic treatment.

4. Experiments

4.1. Dataset

4.1.1. Data preprocessing

In this study, we acquire 2,224 CBCT scans from Shanghai Ninth People's Hospital under consistent acquisition parameters (100 kV source voltage, 0.3 mm voxel size, $468 \times 468 \times 250$ resolution). Each scan undergoes rigid registration to align it to a standardized jaw coordinate system for uniform orientation and field of view, and we apply stringent quality control to remove scans with metal or motion artifacts and incomplete dentition coverage. The selected CBCT scans are processed by an automatic segmentation system (Cui et al., 2022) to generate initial 3D tooth mesh models, which then undergo expert quality assessment where experienced dentists review the results to eliminate incomplete segmentations, correct tooth identification errors, and verify anatomical fidelity. Additionally, the 3D mesh models undergo refinement to ensure fully closed surfaces (i.e., watertight) while eliminating overlaps and intersecting areas. The resulting dataset comprises 1,955 clinically validated high-quality sets of 3D tooth models.

4.1.2. Data split

We divide these 1955 sets of 3D tooth models into two groups: 1507 models for the landmark detection proposed in Section 3.2.1 and the other 448 models for the diffusion model proposed in Section 3.3. (1) For the landmark detection network, we allocate 1507 sets of tooth models, all of which are annotated with tooth landmarks by expert dentists. We use 1207 sets for training our landmark detection algorithm and the remaining 300 sets for testing its performance. (2) For the diffusion model, we allocate the other 448 sets of tooth models, which have undergone manual tooth alignment design by expert dentists, providing us with the ground truth transformation matrices. These 448 sets are initially processed by the pre-trained landmark detection network to automatically identify landmarks on each tooth mesh. Subsequently, 348 sets are used for training the diffusion model, and the other 100 sets are reserved for a comprehensive evaluation of our proposed CLIK-Diffusion.

Notably, all manual annotations are cross-checked by a second dentist. Furthermore, our dataset is significantly larger than those used in previous works (Wei et al., 2020; Wang et al., 2022; Lei et al., 2024; Deng et al., 2024), allowing for a more comprehensive evaluation.

4.1.3. Data demographics

Since our framework is trained and evaluated on the 448 paired pre- and post-orthodontic tooth models, we focus on these 448 tooth models and conduct demographic analysis following ethical approval. The detailed demographic information is shown in Fig. 4.

Specifically, our dataset comprises 448 Chinese participants, with 71.9% females (322 cases) and 28.1% males (126 cases). Ages range from 11 to 48 years, with the majority falling into the 21–30 age group (34.3%, 154 cases), followed by 31–40 age group (31.1%, 139 cases). Younger participants (11–20 years) account for 25.9% (116 cases), while the oldest group (41–48 years) represents 8.7% (39 cases).

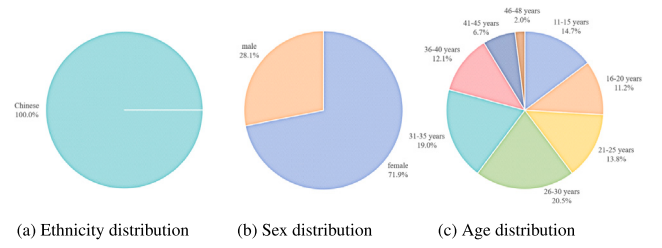


Fig. 4. Demographic information of our dataset.

4.2. Implementation details

Our method is implemented on the PyTorch platform with 1 NVIDIA A100 80 GB GPU. In terms of the landmark detection network, we conduct four PointMLP-based networks of the same architecture for detecting tooth landmarks from incisors, canines, premolars and molars, respectively. We train the landmark detection network for 1000 epochs with the Adam optimizer. The learning rate is initially set to $1e-3$ and decays by 0.5 for every 50 epochs. The batch size is set to 64.

In terms of the diffusion model, we implement a linear variance schedule ranging from $1e-6$ to 0.01 with $T = 2000$ timesteps. The diffusion model undergoes training for 10000 epochs, utilizing the Adam optimizer with a learning rate set to $5e-5$ and batch size set to 48. We set the temperature parameter of Soft-Argmin operator (Goyal et al., 2017; Zhong et al., 2017) to 1000 to enhance contribution of the closest signed distance point in assigning SDF values to query points. The joint loss weights $\lambda_1 = 0.1$, $\lambda_2 = 1e-3$, $\lambda_3 = 0.01$ in Eq. (11), and $\lambda_4 = 1e-3$ in Eq. (9) are optimized through a series of comparative experiments, with each weight iteratively tuned for optimal balance. The denoising module of our diffusion model is based on UNet structure with 1D convolution layers. As shown in Fig. 2, our UNet is designed with four input blocks, each progressively increasing the channel dimension from 64 to 512, and each block contains three residual blocks with dropout (0.2). The middle block utilizes a multi-head self-attention mechanism with 16 heads at a dimension of 512. The four output blocks mirror the input blocks, each containing three residual blocks, decreasing the channel dimension from 512 to 64. The final output is 3-dimensional, representing the 3D coordinates of the predicted tooth landmarks.

4.3. Evaluation metrics

To quantitatively evaluate the performance of our method, we employ four metrics. The rotation error $\Delta_{Rotation}$, measured as an angle in degrees, is computed from the predicted and ground truth rotation matrices R_{pred} and R_{gt} . Inspired by Sattler et al. (2018), Hartley et al. (2013), we compute the rotation error according to the following equation:

$$2 \cos(|\alpha|) = \text{trace} \left(R_{gt}^{-1} R_{pred} \right) - 1, \quad (12)$$

where $\Delta_{Rotation} = |\alpha|$ is the absolute rotation error.

The translation error $\Delta_{Translation}$ measures the difference of translation components between the predicted and ground truth transformation matrices (i.e., T_{pred} and T_{gt}) (Li et al., 2018), mathematically defined as $\frac{1}{3} \|T_{gt} - T_{pred}\|_1$. Moreover, The point cloud error, denoted as $\Delta_{PointCloud}$, is calculated as the average Euclidean distance between corresponding points in the predicted and ground truth point clouds.

Furthermore, to quantify the degree of collision between neighboring teeth, we introduce a new metric $\Delta_{Collision}$ based on Eq. (8). $\Delta_{Collision}$ calculates the average absolute value of the SDF values for all query points which are assigned with negative SDF values, formulated as:

$$\Delta_{Collision} = \frac{\sum_{i=1}^N \sum_{j=1}^{M_i} |\min(\text{SDF}(q_{i,j}), 0)|}{\sum_{i=1}^N \sum_{j=1}^{M_i} \mathbb{1}(\text{SDF}(q_{i,j}) < 0)}, \quad (13)$$

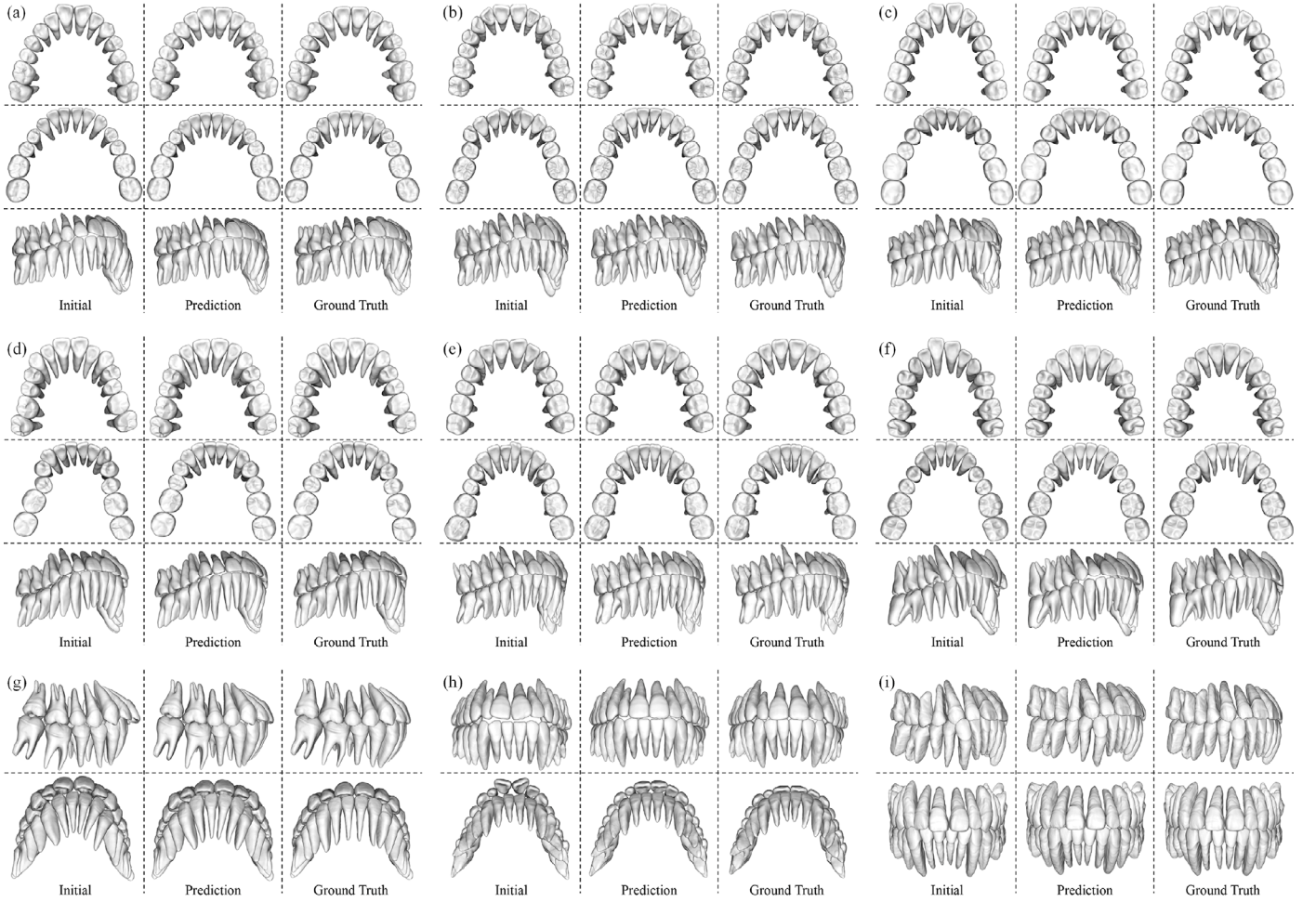


Fig. 5. Visualization of the results of our method. The layout for cases (a, b, c, d, e, f) features a uniform three-row format, where the upper, middle, and lower rows represent the upper teeth, the lower teeth and the side view, respectively. Cases (g, h, i) have a variable layout tailored to the specific case details. For all cases, the three columns from left to right depict the initial dentition, the prediction generated by our method, and the ground truth.

where N is the total number of teeth, M_i is the number of query points for the i th tooth, $SDF(q_{i,j})$ is the SDF value assigned to the j th query point of the i th tooth, and $\mathbb{1}(SDF(q_{i,j}) < 0)$ represents an indicator function that equals 1 if the SDF value is negative, and 0 otherwise. Unlike Eq. (8) which uses a differentiable Soft-Argmin function to identify the closest signed distance point, $\Delta_{Collision}$ utilizes a non-differentiable Argmin (Argmax) operator for higher accuracy. Note that the unit is millimeter (mm) except for $\Delta_{Rotation}$, which is in degree ($^\circ$).

5. Results

We start by visualizing several test cases to demonstrate our method's effectiveness, then compare it quantitatively and qualitatively with existing methods. Subsequently, we conduct extensive ablation studies to validate our proposed hierarchical clinical-inspired constraints and the dentition descriptor technique. Ultimately, user studies are conducted to validate the clinical applicability of our proposed CLIK-Diffusion.

5.1. Landmark detection accuracy

Our landmark detection network, combining PointMLP and heatmap techniques, achieves high accuracy with average distance errors of 0.286 mm for incisors, 0.241 mm for canines, 0.228 mm

for premolars, and 0.250 mm for molars. This precision is crucial for addressing the upcoming tooth alignment task, where precise landmark localization enhances the representation of tooth characteristics. This, in turn, enables the conversion of the problem into a manageable landmark generation task.

5.2. Visualization

As shown in Fig. 5, our method demonstrates its ability to address various malocclusion issues, showcasing its versatility in treating common dental problems. It excels at addressing various orthodontic challenges, including spacing issues (see case (a)), crowding issues (b, c, d, e), overjet (c, f), overbite (g), openbite (h), and crossbite (i). Note that orthodontic cases typically involve a mix of issues rather than a single one. The classification provided here is primarily based on the most significant issue in each case.

Specifically, in the case (a), our method eliminates the significant space between lower teeth, achieving a tighter arrangement. Cases (b) and (d) showcase crowded lower teeth with an irregular dental arch morphology. Our network repositions these individual teeth, achieving a smoother dental arch with a better curvature. Case (f) exemplifies a prevalent malocclusion known as overjet, where our network accurately guides the teeth, particularly the incisors, towards proper alignment. Furthermore, the final three cases (g, h, i) each depict a distinct set of orthodontic challenges that are representative of clinical

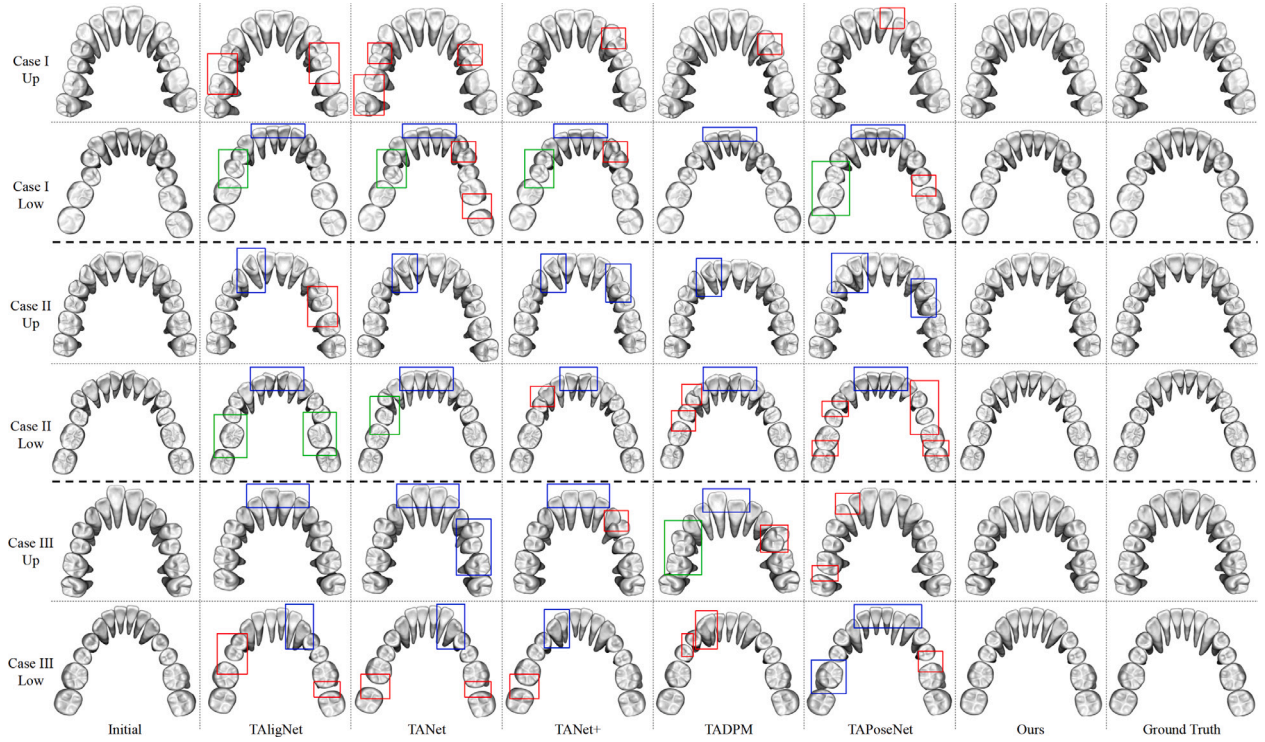


Fig. 6. Visualization of our comparative experiments with the other methods. Here, three cases are shown and each case is represented by two rows: the upper teeth above and the lower teeth below. The red-boxed regions (□) indicate instances of collision or excessive spacing issue between neighboring teeth, while the blue-boxed regions (□) indicate instances of improper orientation of the teeth. The green-boxed regions (□) indicate instances of improper position of the teeth resulting in the irregular dental arch.

Table 1

Quantitative results of comparative experiments with the other methods. The unit is millimeter (mm) except for $\Delta_{Rotation}$, which is in degree (°). The best metrics are shown in **bold**, and the symbol ** denotes p-values less than 0.01 as determined by a paired t-test when compared to ours.

Method	$\Delta_{Rotation} \downarrow$	$\Delta_{Translation} \downarrow$	$\Delta_{PointCloud} \downarrow$	$\Delta_{Collision} \downarrow$
TAligNet (Lingchen et al., 2020)	12.303 ± 2.213**	2.555 ± 0.516**	2.931 ± 0.790**	0.503 ± 0.151**
TANet (Wei et al., 2020)	10.089 ± 1.988**	2.095 ± 0.465**	2.907 ± 0.706**	0.544 ± 0.135**
TANet+ (Wang et al., 2022)	9.965 ± 2.147**	2.016 ± 0.463**	2.076 ± 0.693**	0.301 ± 0.139**
TADPM (Lei et al., 2024)	9.928 ± 2.537**	2.210 ± 0.542**	2.426 ± 0.655**	0.306 ± 0.142**
TAPoseNet (Deng et al., 2024)	9.331 ± 1.129**	1.775 ± 0.310**	1.864 ± 0.543**	0.345 ± 0.154**
Ours	6.669 ± 1.577	1.305 ± 0.344	1.415 ± 0.529	0.213 ± 0.130

scenarios. The overbite issues depict a scenario where the upper teeth extend beyond the lower teeth to an excessive degree when a person bites down his teeth. In the case (g), the upper incisors protrude excessively over the lower teeth, disrupting the natural occlusion. Our method manages to reposition the incisors inward and rotate them to the proper orientation simultaneously. Case (h) slightly suffers from a rare type of malocclusion, clinically termed as openbite, namely some of the teeth do not meet when biting down. In this instance, the four upper incisors fail to meet their corresponding lower incisors, as depicted in case (h). Our method manages to move these incisors closer together to meet the need of normal occlusion. Moreover, case (i) exhibits a slight crossbite issue, which means that the upper teeth locate inside the lower teeth when biting down. Specifically, the left canine of the lower teeth extends slightly outside the corresponding upper canine. Our method meticulously repositions the canine to achieve a more ideal occlusion.

In conclusion, the results achieved by our CLIK-Diffusion, as shown in Fig. 5, not only enhances the aesthetic appearance but also promotes proper bite function and reduces risk of dental complications.

5.3. Comparison

To verify the effectiveness of our proposed method, we implement several recent state-of-the-art methods in the area of tooth alignment

for comparison: TAligNet (Lingchen et al., 2020), TANet (Wei et al., 2020), TANet+ (Wang et al., 2022), TADPM (Lei et al., 2024) and TAPoseNet (Deng et al., 2024). The quantitative experimental results are shown in Table 1. Compared to TAligNet, our method achieves significant improvement across all metrics, with the improvement in all metrics exceeding 45%. TANet, recognized as the first method to automatically address the tooth alignment problem, employs a PointNet encoder for both jaw-level and tooth-level feature extraction, followed by a graph neural network to learn the transformation represented in axis-angle format. TANet+, a journal extension of TANet, introduces partial tooth landmarks as additional constraints to make teeth normally oriented, resulting in a better improvement over TANet. However, even compared to the enhanced TANet+, our method achieves substantial improvement across all metrics, with remarkable enhancements ranging from 26% to 35%. TADPM demonstrates comparable tooth alignment capability to TANet+, though it exhibits a little weaker performance in repositioning teeth to their correct positions, resulting in a relatively higher error in $\Delta_{Translation}$ compared to TANet+. Moreover, TAPoseNet achieves better performance compared with other existing methods, with nearly all metrics showing improvement over other methods except ours. This is particularly evident in both $\Delta_{Translation}$ and $\Delta_{PointCloud}$ metrics, which highlight its relatively strong ability to reposition teeth. Building upon these comparative methods, our proposed method achieves better performance across all metrics. Notably,

compared to the more recent TAPoseNet, our method still achieves significant improvement, with a 28.5% improvement in $\Delta_{Rotation}$, 26.5% in $\Delta_{Translation}$, 24.1% in $\Delta_{PointCloud}$ and 38.3% in $\Delta_{Collision}$.

To further demonstrate the advantage of our method, we present a visual comparison of three typical cases in Fig. 6. Overall, the visualization results clearly indicate that all existing methods suffer from several significant limitations. As highlighted in the red-boxed regions, all methods, except ours, fail to adequately prevent unwanted collision and excessive spacing issue between neighboring teeth for both upper and lower teeth. Specifically, TAPoseNet generates dentitions with relatively fewer collisions. However, this improvement comes at the cost of introducing excessive inter-tooth space as a compensatory mechanism for collision avoidance. In addition, as shown in the blue-boxed regions, they fall short in predicting proper orientation for each tooth, particularly for incisors. Furthermore, the green-boxed regions reveal an intuitive limitation of existing methods, and then demonstrate the effectiveness of our proposed clinical knowledge-informed constraints. For example, in the case I, we observe that the lower dental arch exhibits irregular morphology, with the premolar receding inward and other teeth being crowded. Our method effectively predicts an ideal dentition, positioning the premolar and orienting the other teeth appropriately. In contrast, TANet fails to reposition the premolar, whereas TALigNet, TANet+ and TAPoseNet all manage to adjust it to a relatively correct position, but resulting in a collision with its neighboring tooth. In summary, our method has visually shown exceptional efficacy compared with existing methods, particularly in the realm of positioning and orientation correction for each tooth. This advancement significantly contributes to mitigating orthodontic issues (e.g., crowding and overjet issues). Nevertheless, our method encounters certain limitations. Upon comparison with the ground truth annotations established by expert dentists, our model exhibits a degree of inaccuracy in the inter-tooth areas, occasionally resulting in minor collision or slight spacing issue between teeth.

Both quantitative results and qualitative visualizations clearly indicate that all existing methods struggle to predict an accurate transformation for each tooth. This challenge partly arises from the fact that all approaches rely heavily on encoder-based feature extraction, both globally and locally. This formulation inevitably leads to significant loss of geometric information during encoding, especially about the key information for treatment (e.g., tooth tip points), making the network performance highly dependent on the training quality of the encoder. More importantly, while other methods frame the tooth alignment problem as a transformation regression task, our approach considers it a landmark coordinate generation task. This landmark-based formulation creates a natural pathway for incorporating clinical knowledge into the network. In orthodontic treatment, tooth movement follows well-established clinical principles. While existing methods that directly predict post-orthodontic transformations lack the flexibility to integrate these clinical rules, our framework enables explicit encoding of clinical knowledge through carefully designed hierarchical loss functions.

5.4. External validation

The first and only public dataset in this domain (Wang et al., 2024) comprises 1,060 pairs of pre- and post-orthodontic tooth models collected from Beijing Stomatological Hospital. This dataset, originating from the same research team that develops TADPM (Lei et al., 2024), is divided into 810 pairs for training and 250 for testing. Notably, there is a domain gap between the tooth models in this public dataset and those used in our study, as well as those employed by other existing methods. Specifically, unlike the data used by existing methods, this dataset includes gingival structures, which they claim preserving the integrity of contact surfaces. Moreover, this dataset lacks root structures due to the intrinsic limitations of optical intraoral scanning, whereas our dataset is derived from CBCT images and therefore contains complete root information.

As for the implementation, to accommodate the aforementioned differences between datasets, we retrain our landmark detection network using a subset of 100 tooth models from the training set (80 for training, 20 for validation), with the same panel of dentists annotating landmarks. Following our training strategy, our landmark detection network consistently achieves high accuracy with average distance errors of 0.189 mm for incisors, 0.183 mm for canines, 0.166 mm for premolars, and 0.178 mm for molars.

Regarding comparative methods, we implement TANet, TANet+, and TADPM on this external dataset, excluding TALigNet and TAPoseNet. We omit TALigNet because its simplistic architecture yields relatively trivial results. Implementing TAPoseNet on this external dataset is challenging due to its tooth pose estimation module. Given the aforementioned domain gap between the external dataset and the standard dataset, no existing methods currently support automatic tooth pose estimation on such data, and manual annotation would be prohibitively time-consuming.

Quantitative results on the external dataset are presented in Table 2. Our method not only significantly outperforms TANet and TANet+ by a considerable margin, but also achieves superior performance compared to more recent method, TADPM, particularly in $\Delta_{Rotation}$ and $\Delta_{Translation}$ (with approximately 30% improvement). This highlights our method's strong capability in accurately predicting the correct orientation and position of post-orthodontic teeth. Although the improvement over TADPM in $\Delta_{PointCloud}$ and $\Delta_{Collision}$ is relatively modest, it highlights our method's enhanced capability in overall tooth alignment and collision reduction. Moreover, the consistently lower standard deviations across all metrics further indicate that our method exhibits greater robustness when applied to diverse dental cases.

Visualization of comparative results in Fig. 7 reveals distinct performances among these methods performed on the external dataset. TANet demonstrates the weakest visual performance, partially correcting tooth orientation in simpler cases (e.g., upper incisors in Cases I and III, marked by blue boxes), but failing in more challenging scenarios (e.g., lower incisors in Cases I, II and III). Additionally, it also produces suboptimal position predictions (see green-boxed regions) and frequent collision and spacing issues (see red-boxed regions). TANet+ improves upon TANet with better orientation constraints, achieving more accurate orientation predictions for cases like lower incisors in Cases I and III. However, it still struggles with position predictions and exhibits collision or spacing issues. While TADPM generates dental arches relatively close to ground truth, detailed imperfections remain. In particular, the orientations of some teeth still fall short of clinical acceptability. This is consistent with the quantitative results as TADPM's $\Delta_{Rotation}$ remains relatively high, while other metrics are relatively low. In contrast, our method outperforms other methods, achieving superior accuracy in tooth orientation prediction, position correction, and optimal inter-tooth relationships with minimal collision and spacing issues.

Both quantitative and visual results indicate our method's superior performance on external datasets. Despite the significant domain gap (including gingiva but lacking root), our approach remains robust. This robustness primarily stems from our landmark-based representation, which effectively captures tooth position, orientation and shape without relying heavily on detailed format of tooth models.

5.5. Ablation study

5.5.1. Evaluation of hierarchical constraints

We conduct extensive experiments to demonstrate the effectiveness of the knowledge-informed hierarchical constraints, including the dental-arch level $\mathcal{L}_{Arch}$, inter-tooth level $\mathcal{L}_{Inter}$ and individual-tooth level $\mathcal{L}_{Individual}$. Table 3 and Fig. 8 present the results of four experiments, where we incrementally add constraints from the global level to the local level. The results demonstrate that the progressive incorporation of constraints leads to a corresponding improvement in

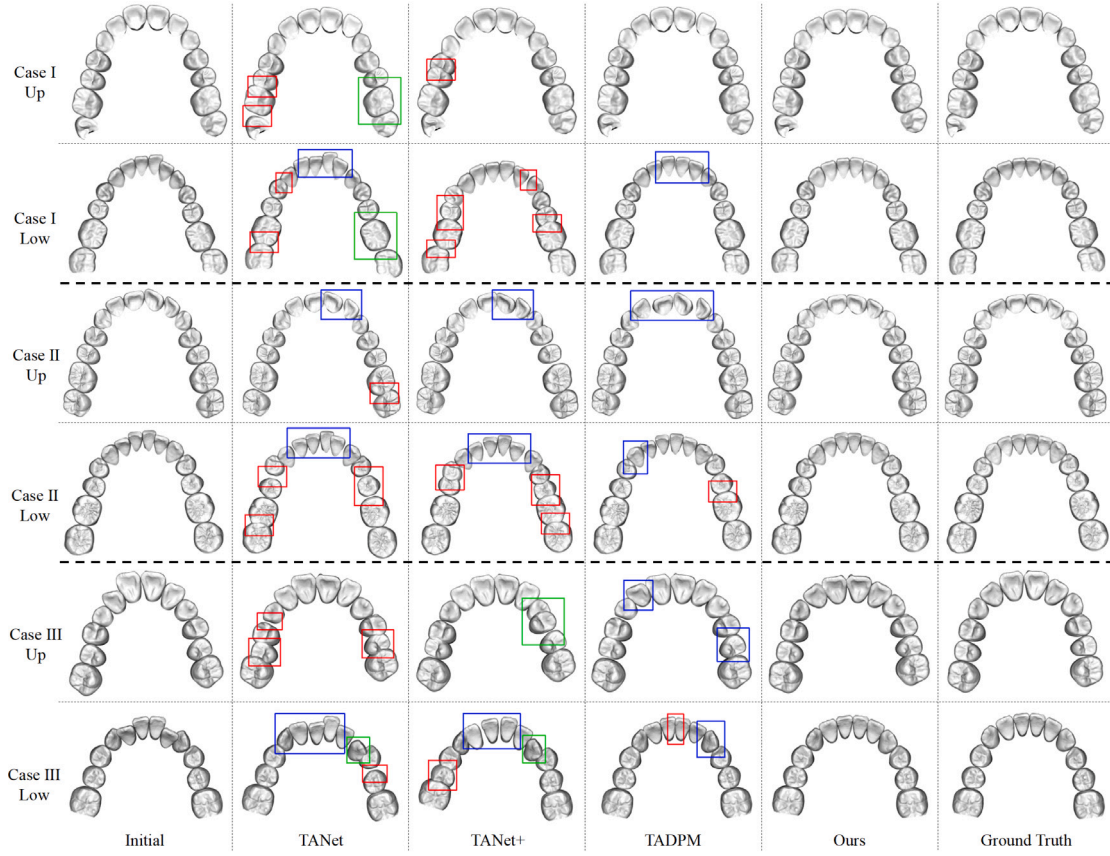


Fig. 7. Visualization results on the external dataset (Wang et al., 2024) compared with other methods. Here, three cases are shown and each case is represented by two rows: the upper teeth above and the lower teeth below. The meanings of color-coded boxed regions are consistent with those defined in Fig. 6.

Table 2

Quantitative results on the external dataset (Wang et al., 2024) compared with other methods. The unit is millimeter (mm) except for $\Delta_{Rotation}$, which is in degree ($^{\circ}$). The best metrics are shown in **bold**, and the symbol ** denotes p-values less than 0.01 as determined by a paired t-test when compared to ours.

Method	$\Delta_{Rotation} \downarrow$	$\Delta_{Translation} \downarrow$	$\Delta_{PointCloud} \downarrow$	$\Delta_{Collision} \downarrow$
TANet (Wei et al., 2020)	11.685 $\pm$ 3.189**	2.505 $\pm$ 0.624**	3.593 $\pm$ 1.307**	0.609 $\pm$ 0.205**
TANet+ (Wang et al., 2022)	10.878 $\pm$ 3.028**	2.173 $\pm$ 0.551**	2.878 $\pm$ 1.086**	0.696 $\pm$ 0.157**
TADPM (Lei et al., 2024)	9.807 $\pm$ 3.782**	1.765 $\pm$ 0.678**	1.543 $\pm$ 0.722**	0.194 $\pm$ 0.129**
Ours	6.688 $\pm$ 2.561	1.199 $\pm$ 0.441	1.295 $\pm$ 0.678	0.168 $\pm$ 0.128

each metric. (1) Specifically, $\mathcal{L}_{Arch}$, acting as a constraint on the global feature of the overall morphology of the dental arch, yields the most significant improvement across all metrics except $\Delta_{Rotation}$, which still shows a substantial, albeit not the largest, improvement (See E2). This is visually apparent in Fig. 8, where the shape and size of the dental arch improve significantly once $\mathcal{L}_{Arch}$ is incorporated (E2), compared to the smaller arch produced by $\mathcal{L}_{Base}$ alone (E1). (2) On the other hand, $\mathcal{L}_{Inter}$, which constrains the position of neighboring teeth, manages to reduce unnecessary collisions between neighboring teeth, as seen in the zoomed-in area of E2 and E3 in Fig. 8. Given the small weight assigned to $\mathcal{L}_{Inter}$, the improvement in metrics is relatively minor. (3) Furthermore, $\mathcal{L}_{Individual}$, which constrains the orientation of each individual tooth, leads to relatively large improvement in all metrics, especially in $\Delta_{Rotation}$ (E4), which aligns with its intended purpose. With all hierarchical constraints incorporated, along with the inherent loss of diffusion model $\mathcal{L}_{Base}$, our CLIK-Diffusion can generate a well-aligned dentition with all teeth correctly oriented and positioned. The results of our ablation study demonstrate that each knowledge-informed constraint introduced into the diffusion model is clinically practical for orthodontic treatment and contributes to significant improvement in the effectiveness of our proposed CLIK-Diffusion.

5.5.2. Evaluation of dentition descriptor

A sufficient number of landmarks can fairly accurately represent the position, orientation, and shape of each individual tooth. However, to enable our model to learn the tooth features more adequately, particularly their shapes, we employ a technique called dentition descriptor to sample vertices surrounding each landmark, as detailed in Section 3.2.2. To validate the effectiveness of the dentition descriptor, we conduct ablation experiments by training our framework without the dentition descriptor and the associated three-layer MLP component. These ablation experiments are constructed based on the configuration of experiments E1 and E4, which are defined in Table 3. The results of our ablation experiments, presented in Table 4, demonstrate statistically significant improvement across almost all metrics, despite the relatively small enhancements. These findings suggest that the key landmarks used in our framework already provide essential information from global dentition structure to local tooth shapes, achieving promising performance. And the dentition descriptor can further refine our model, capable of leveraging subtle yet effective adjustments in the learning process.

Table 3

Ablation study of the hierarchical knowledge-informed constraints in our CLIK-Diffusion. The unit is millimeter (mm) except for $\Delta_{Rotation}$, which is in degree ($^{\circ}$). The best metrics are shown in **bold**, and the symbols * and ** denote p-values less than 0.05 and 0.01, respectively, as determined by a paired t-test when compared to the preceding experimental condition.

	$\mathcal{L}_{Base}$	$\mathcal{L}_{Arch}$	$\mathcal{L}_{Inter}$	$\mathcal{L}_{Individual}$	$\Delta_{Rotation} \downarrow$	$\Delta_{Translation} \downarrow$	$\Delta_{PointCloud} \downarrow$	$\Delta_{Collision} \downarrow$
E1	✓				8.484 $\pm$ 1.758	2.647 $\pm$ 0.373	4.102 $\pm$ 0.434	0.597 $\pm$ 0.078
E2	✓	✓			7.885 $\pm$ 1.715**	1.841 $\pm$ 0.362**	2.299 $\pm$ 0.454**	0.400 $\pm$ 0.106**
E3	✓	✓	✓		7.740 $\pm$ 1.676*	1.691 $\pm$ 0.371**	1.972 $\pm$ 0.461**	0.343 $\pm$ 0.108**
E4	✓	✓	✓	✓	6.669 $\pm$ 1.577**	1.305 $\pm$ 0.344**	1.415 $\pm$ 0.529**	0.213 $\pm$ 0.130**

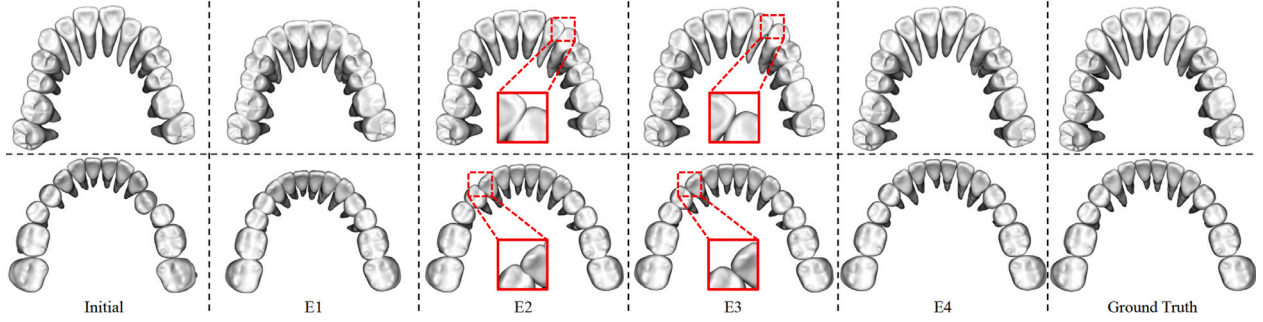


Fig. 8. Visualization of our ablation study of the hierarchical constraints in our CLIK-Diffusion. The upper row represents the upper teeth and the lower row represents the lower teeth. E1 to E4 are four experiments consistent with Table 3, denoting the hierarchical constraints are progressively incorporated into the diffusion model.

Table 4

Ablation study of the dentition descriptor technique. The unit and symbols *, ** are consistent with Table 3.

	$\Delta_{Rotation} \downarrow$	$\Delta_{Translation} \downarrow$	$\Delta_{PointCloud} \downarrow$	$\Delta_{Collision} \downarrow$
E1 w/o descriptor	11.705 $\pm$ 1.565	2.845 $\pm$ 0.384	4.231 $\pm$ 0.674	0.412 $\pm$ 0.089
E1 w/ descriptor	8.484 $\pm$ 1.758**	2.647 $\pm$ 0.373**	4.102 $\pm$ 0.434*	0.597 $\pm$ 0.078**
E4 w/o descriptor	6.801 $\pm$ 1.616	1.404 $\pm$ 0.356	1.541 $\pm$ 0.528	0.245 $\pm$ 0.108
E4 w/ descriptor	6.669 $\pm$ 1.577*	1.305 $\pm$ 0.344**	1.415 $\pm$ 0.529**	0.213 $\pm$ 0.130**

5.5.3. Choice of backbone

Our comparative experiments against existing methods, along with external validation studies, have demonstrated that formulating tooth alignment as a landmark generation task achieves superior performance compared to conventional transformation matrix regression tasks. Furthermore, our landmark-based framework inherently facilitates the integration of clinical knowledge, which has been demonstrated through previous ablation studies to significantly enhance the practical applicability.

Given that our method formulates tooth alignment as a generation problem, we employ the diffusion model for its established generative capabilities. We conduct comprehensive comparative experiments with Pix2Pix GAN (Isola et al., 2017) and conditional VAE (Sohn et al., 2015) under identical training protocols, aimed to confirm the superiority of our diffusion-based approach.

Quantitative results are presented in Table 5. Our diffusion-based approach significantly outperforms cVAE-based and cGAN-based methods in tooth alignment prediction, reducing $\Delta_{Rotation}$ and $\Delta_{Translation}$ by approximately 20%. The lower standard deviations further confirm its robustness across diverse cases. While all methods show comparable $\Delta_{PointCloud}$ metrics, our approach maintains statistically significant.

Visual results are shown in Fig. 9. Both cVAE- and cGAN-based methods effectively correct moderately misaligned teeth (e.g., upper incisors in Cases I and III), but struggle with severe misalignment (e.g., lower incisors in Case II), as evidenced by frequent blue-boxed regions where orientation prediction fails. In terms of position correction, both methods perform well, exhibiting fewer green-boxed regions. Additionally, they relatively mitigate collision between neighboring teeth. This phenomenon reflects the trade-off between precise alignment and collision avoidance, as an angled tooth occupies less space than a properly aligned one, thereby inherently reducing collision risks.

Notably, although underperforming ours, both cVAE- and cGAN-based methods still outperform existing methods in Table 1 and Fig. 6.

This validates the superiority of our landmark generation formulation over transformation regression formulation and highlights the crucial role of clinical knowledge-informed hierarchical constraints in tooth alignment.

5.6. User study

To further demonstrate the clinical applicability of our method, we conduct a user study to assess the alignment degree of our predicted dentition. Specifically, we invite 13 orthodontists to participate in our study, including 4 expert orthodontists with more than 5 years of experience in the field and 9 junior orthodontists with less than 5 years of experience. For each orthodontist, we randomly select 18 pairs of cases from different patients. Each pair consists of one ground truth dentition which is pre-annotated by other experienced orthodontists, and one predicted dentition generated by our method. The orthodontists are requested to grade the alignment degree of the two given dentitions on a scale of 0 to 10, with higher scores indicating better alignment. In addition, after grading, the orthodontists are asked to identify which dentition they believe to be the ground truth, simultaneously providing a confidence score ranging from 0 to 5 in ascending order. A score of 0 indicates they cannot distinguish between the two at all, while a score of 5 indicates they are completely certain of their choice. If the participant's choice is incorrect, the confidence score is recorded as negative (i.e., multiplied by -1). To facilitate their evaluations, we encourage the orthodontists to view the 3D tooth models in mesh format using professional software, allowing them to rotate and scale the meshes arbitrarily.

Table 6 provides a comparison of alignment scores for the predicted and ground truth dentition, evaluated by both expert and junior orthodontists. The alignment score of the ground truth dentition represents the upper limit of the alignment degree as perceived by orthodontists. Therefore, the closer the predicted dentition's alignment

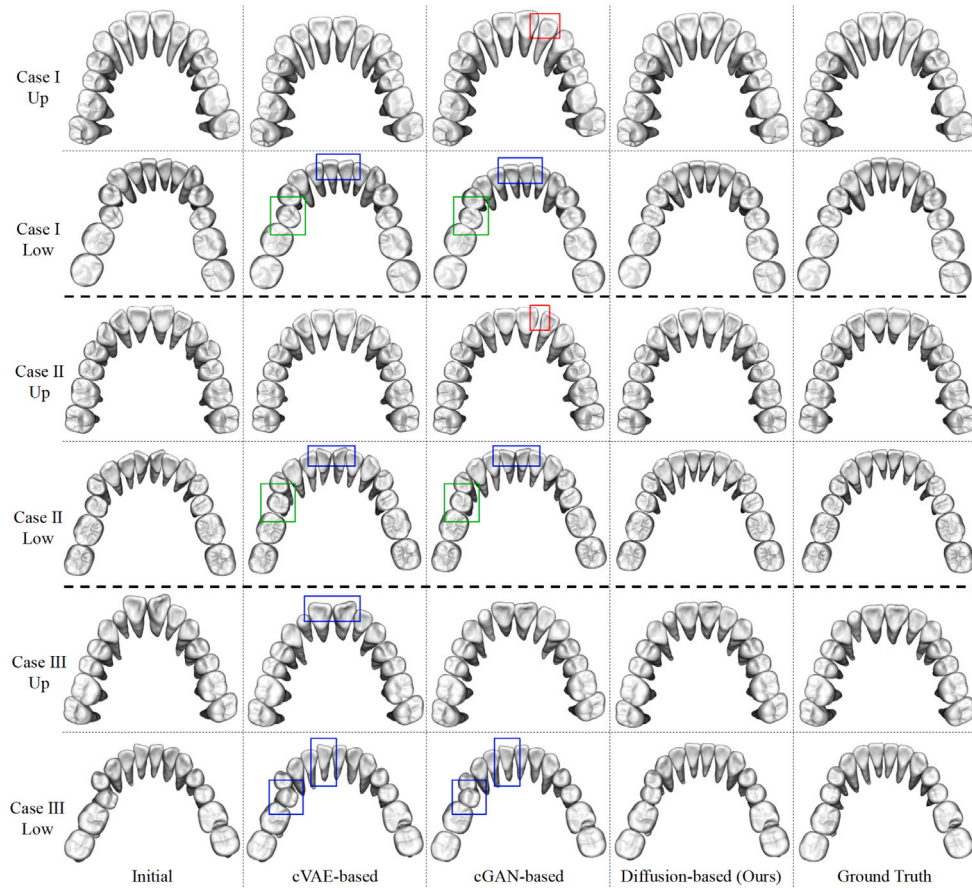


Fig. 9. Visualization results compared with cVAE-based and cGAN-based backbones. Here, three cases are shown and each case is represented by two rows: the upper teeth above and the lower teeth below. The first two cases correspond to the same patients shown in the first two cases of Fig. 6. The meanings of color-coded boxed regions are consistent with those defined in Fig. 6.

Table 5

Quantitative results compared with cVAE-based and cGAN-based backbones. The unit is millimeter (mm) except for $\Delta_{Rotation}$, which is in degree ($^{\circ}$). The best metrics are shown in **bold**, and the symbols * and ** denote p-values less than 0.05 and 0.01, respectively, as determined by a paired t-test when compared to ours.

Method	$\Delta_{Rotation} \downarrow$	$\Delta_{Translation} \downarrow$	$\Delta_{PointCloud} \downarrow$	$\Delta_{Collision} \downarrow$
cVAE-based (Sohn et al., 2015)	9.175 $\pm$ 2.503**	1.795 $\pm$ 0.507**	1.490 $\pm$ 0.593**	0.110 $\pm$ 0.055**
cGAN-based (Isola et al., 2017)	8.682 $\pm$ 1.785**	1.648 $\pm$ 0.384**	1.508 $\pm$ 0.465**	0.194 $\pm$ 0.099*
Diffusion-based (Ours)	6.669 $\pm$ 1.577	1.305 $\pm$ 0.344	1.415 $\pm$ 0.529	0.213 $\pm$ 0.130

score is to this value, the more effectively our method demonstrates its efficacy. Both expert and junior orthodontists give high ratings to the predicted dentition, suggesting that our method is close to or potentially suitable for clinical application. Moreover, the smaller variance in the predicted dentition's scores indicates that results of our method are aesthetically satisfactory to a wider range of evaluators, thus enhancing its clinical reliability.

Fig. 10 illustrates the distribution of confidence scores assigned by expert and junior orthodontists when identifying the ground truth dentition. For junior orthodontists, the balance between positive and negative confidence scores suggests that our method makes it challenging for them to distinguish between the predicted results and the ground truth. This demonstrates that our method has achieved a certain level of clinical practicality, particularly in assisting junior orthodontists. In contrast, expert orthodontists have more positive confidence scores, indicating they can more easily differentiate between the ground truth and our predicted results. This suggests that our method can be further optimized for improved performance.

6. Discussion

6.1. Advantages of the proposed model

In the realm of digital dentistry, tooth alignment is an essential stage, aiming to anticipate an ideal dental arrangement from a set of misaligned teeth. The development of a clinically viable method for automatic tooth alignment is crucial, and it has captivated researchers both in clinical and technical fields. Nonetheless, current methods consistently fall short of delivering satisfactory outcomes. A commonality among these methods is their heavy reliance on encoders to derive features from the input tooth models. Moreover, these methods, without exception, formulate the tooth alignment problem as a transformation regression task.

To tackle aforementioned issues, we introduce an innovative clinical knowledge-informed diffusion model, leveraging the recently demonstrated superiority of generation across various domains. We reframe the intricate problem of tooth alignment into the more tractable issue

Table 6

Results of our user study on alignment scores. We calculate the average alignment scores graded by all participants for all cases, along with the corresponding standard deviations.

	Expert Orthodontist	Junior Orthodontist	Total
Predicted Dentition	7.486 $\pm$ 1.221	7.833 $\pm$ 1.029	7.727 $\pm$ 1.101
Ground Truth Dentition	8.056 $\pm$ 1.255	7.994 $\pm$ 1.139	8.013 $\pm$ 1.174

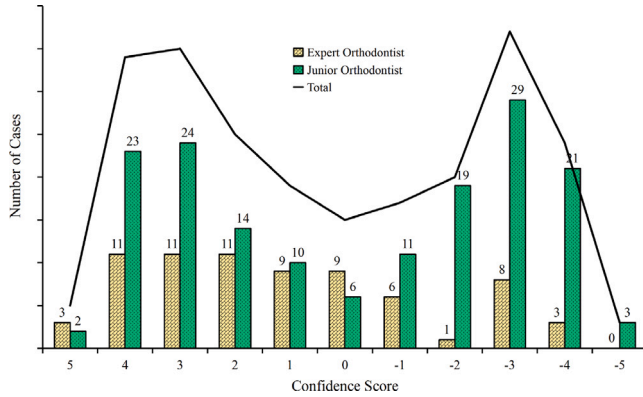


Fig. 10. Distribution of confidence scores for identifying the ground truth dentition. The yellow bars represent the confidence scores assigned by expert orthodontists, while the green bars correspond to those from junior orthodontists. The black line indicates the total number of cases for each confidence score. Positive scores reflect correct identification, with higher values indicating greater certainty, while negative scores reflect incorrect identification.

of landmark transformation, subsequently simplifying it further into a task of generating landmark coordinates, which is more intuitive than predicting the transformation matrices of individual teeth by neural network. Particularly, our approach incorporates hierarchical clinical-inspired constraints and transforms these constraints into specifically designed losses within our CLIK-Diffusion. These constraints stem from practical clinical knowledge, embodying the experience of expert dentists and adhering to established clinical guidelines in dentistry. Notably, our method pioneers the use of 3D tooth models segmented from CBCT images, a departure from the conventional use of intraoral scans by existing methods. Compared with intraoral scans, CBCT images provide a distinct advantage by producing watertight meshes at the interfaces of neighboring teeth, which is vital in clinical practice for the accurate assessment of contact relationships.

6.2. Performances in challenging cases

To evaluate our method's performance in challenging scenarios, particularly cases involving missing teeth, we show four representative cases. As shown in Fig. 11, these include Cases I and II from our internal dataset, along with Cases III and IV from the external dataset (Wang et al., 2024). Cases I and III demonstrate scenarios where the space created by missing teeth must be actively redistributed to alleviate crowding (typically occurring in tooth extraction cases). In contrast, Cases II and IV represent situations where the space requires no redistribution (commonly seen in congenital tooth agenesis or cases requiring future restorative treatment). Overall, our method demonstrates relatively effective performance in both scenarios, in some instances producing even better results compared to inaccurate ground truth (see the blue-boxed region in Case II). Nevertheless, some limitations remain. For instance, in Cases I and III, the necessity for extensive tooth movements to fill the space leads to occasional collision or spacing issues (see the red-boxed regions). We anticipate that increasing the quantity and diversity of such cases in our training dataset would help resolve these challenges.

6.3. Generalizability to different populations

Our method incorporates hierarchical constraints informed by clinical knowledge that is universally applicable across populations, which supports generalizability across different populations. Specifically, both the inter-tooth level and individual-tooth level constraints show minimal dependence on demographic attributes such as ethnicity, sex, or age, thereby enabling broad applicability. In contrast, the dental-arch level constraint, which mainly regulates the size of the dental arch, does exhibit some correlation with demographic variables. Although our dataset includes only individuals of Chinese ethnicity, it covers a diverse and relatively balanced distribution of sex and age. Relevant research (Bishara et al., 1997; Azlan et al., 2019; Lombardo et al., 2015) indicates that age-related and inter-individual variations in dental arch size may be comparable to or even greater than those observed across different ethnicities. Therefore, the diversity in sex- and age-related dental arch in our data may reflect the potential applicability of our method to individuals from other ethnic backgrounds. Nonetheless, we acknowledge that incorporating a more ethnically diverse dataset, where available, would further strengthen the empirical evidence for the robustness of our method.

6.4. Future work

While our current method demonstrates promising results, addressing the critical constraint of avoiding contact between teeth and alveolar bone during clinical tooth movement remains an important challenge. In the future, we aim to develop robust strategies to integrate such anatomical constraints into our network, thereby enhancing clinical applicability of our method.

Moreover, while our current dataset already significantly exceeds those claimed in existing methods, we aim to further expand its diversity and scale, if available, by including a broader range of sex, age, and ethnic demographics, as well as a wider variety of malocclusion types. This enhanced dataset would enable a more comprehensive validation of the applicability and generalizability of our proposed CLIK-Diffusion.

7. Conclusion

In this paper, we introduce a novel clinical knowledge-informed diffusion model for tooth alignment, leveraging hierarchical clinical-inspired constraints. We pioneer the conversion of the complex tooth alignment challenge into a more manageable landmark transformation problem, and further, into a landmark coordinate generation task. Our experimental results demonstrate the effectiveness of this innovative problem formulation compared to existing methods. Additionally, we are the first to comprehensively implement the hierarchical constraints, spanning from the global dental-arch level through the inter-tooth level to the local individual-tooth level, which are crucial in practical applications. These clinical knowledge-informed constraints are designed to address specific aspects of the problem while simultaneously collaborating to enhance the overall effectiveness of our framework. In conclusion, our method can address various types of malocclusion issues, making our method highly reliable and potentially applicable in clinical practice.

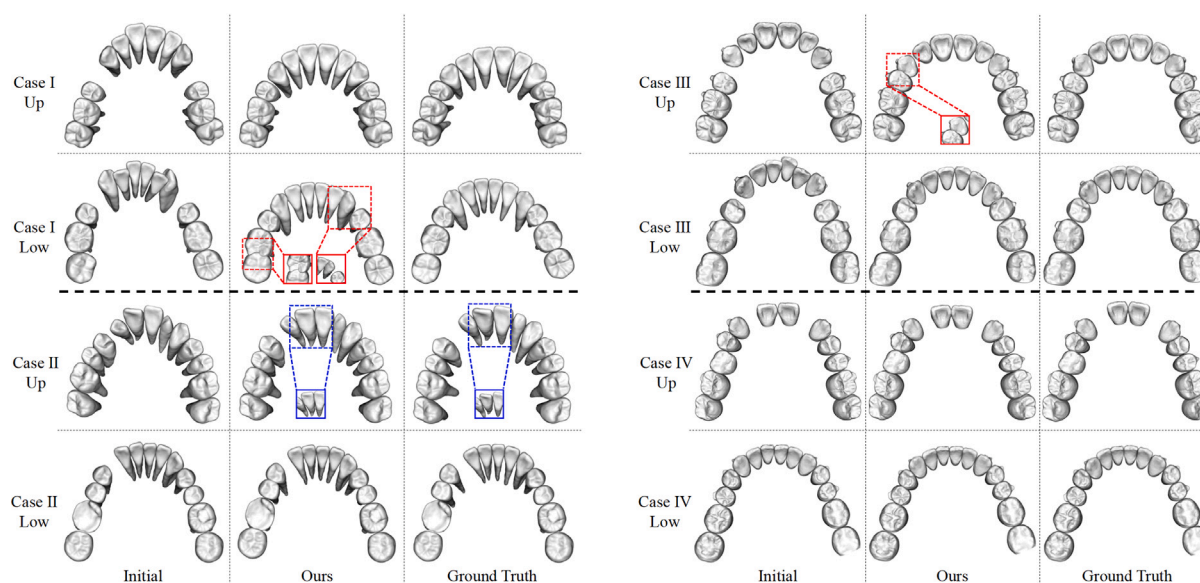


Fig. 11. Visualization results of cases with missing teeth. Here we show four test cases, two of which (Cases I and II) are from our own dataset and the other two (Cases III and IV) are from the external dataset (Wang et al., 2024). Each case is represented by two rows: the upper teeth above and the lower teeth below. The red-boxed zoom-in region indicates the contact relationship between neighboring teeth, while the blue-boxed zoom-in region shows the orientation of the teeth.

CRedit authorship contribution statement

Yulong Dou: Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Han Wu:** Writing – review & editing, Investigation. **Changjian Li:** Writing – review & editing. **Chen Wang:** Validation. **Tong Yang:** Data curation. **Min Zhu:** Validation, Investigation, Conceptualization. **Dinggong Shen:** Writing – review & editing. **Zhiming Cui:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Co-author serves as a senior editor for Medical Image Analysis - Dinggang Shen. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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